To determine input characteristics, the collector base voltage V_{CE} is kept constant at zero and base current I_B is increased from zero by increasing V_{BE}. This is repeated for higher fixed values of V_{CE}.

A curve is drawn between base current and base emitter voltage at constant collector base voltage is shown in figure 5.17. Here the base width decreases. So curve moves right as V_{CE} increases.

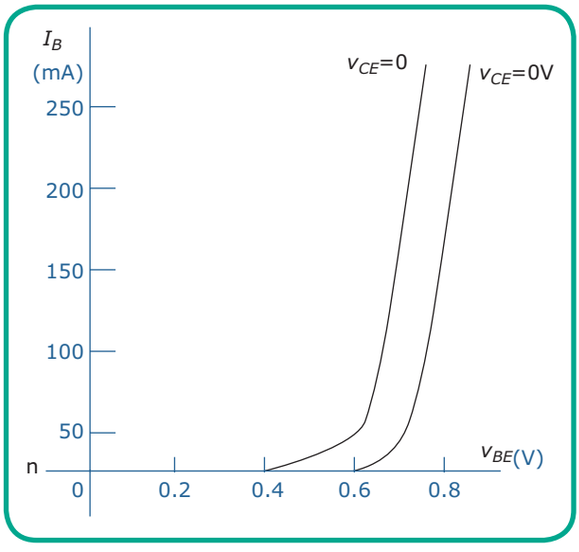


Figure 5.17 Input characteristics

Output Characteristics

It is defined as the characteristic curve drawn between output voltage to output current whereas input current is constant.

To determine output characteristics, the base current I_B is kept constant at zero and collector current Ic is increased from zero by increasing V_{CE}. This is repeated for higher fixed values of I_B.

From the characteristic it is seen that for a constant value of I_B, Ic is independent of V_{CB} and the curves are parallel to the axis of V_{CE} as shown in the figure 5.18.

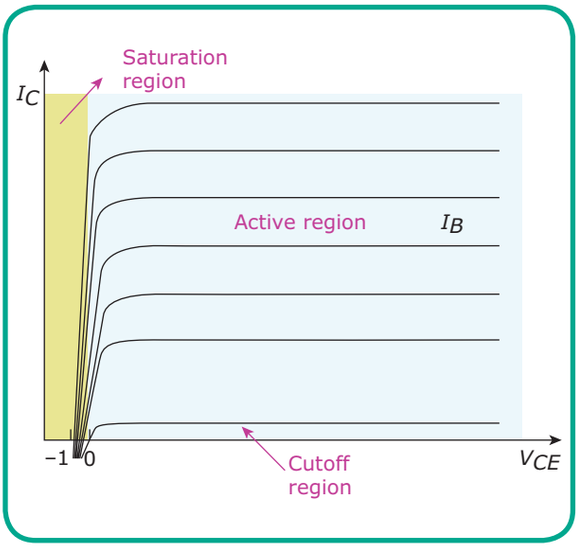


Figure 5.18 Output characteristics

Base current amplification factor (β).

It is the ratio of output current to input current. In common emitter connection, the input current is the base current I_B and output current is the collector current I_C.

The ratio of change in collector current to the change in base current at constant collector emitter voltage V_{CE} is known as current amplification factor i.e.

$$\beta = \frac{\Delta I_C}{\Delta I_B} \text{ at constant } V_{CE}$$

5.9.3 Common Collector Configuration

In common collector configuration circuit is shown in figure 5.19. Here collector is grounded and it is used as the common terminal for both input and output. It is also called as grounded collector configuration. Base is used as an input terminal whereas emitter is the output terminal.

Input Characteristics

It is defined as the characteristic curve drawn between input voltage to input current whereas output voltage is constant.

ELEC_EQUIP_Chapter05.indd 108 11/10/2020 11:41:47 AM